MIL-B-25870A(USAF)
20 July 1972
SUPERSEDING
MIL-B-25870(USAF)
13 March 1958

MILITARY SPECIFICATION

BEARINGS SINTERED METAL, PROCESSING OF

1. SCOPE

1.1 This specification presents requirements for the processing and installation of sintered bushings and parts for use as bearings.

* 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
VV-L-800	Lubricating Oil, General Purpose, Preservative, (Water-displacing, low temperature)

Military

MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-B-5687	Bearings, Sleeve, Washers, Thrust, Sintered, Metal Powder, Oil-impregnated

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B46.1 Surface Texture, Surface Roughness, Waviness and Lay

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:25Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, New York 10017.)

3. REQUIREMENTS

3.1 Material.

3.1.1 This specification applies to the processing of sintered material as described in MIL-B-5687 or similar materials for use as bushing type bearings.

* 3.1.2 Unless otherwise specified, the lubricant used to impregnate or reimpregnate the bearings shall be in accordance with VV-L-800.

3.1.3 Unless otherwise specified, bearings processed from sintered materials as specified herein shall conform to the requirements of MIL-B-5687.

3.2 General requirements. The material shall be protected from surface injury and contamination during all stages of handling, processing and storage. The successful use of sintered metal oil impregnated bearings depends upon the maintenance of uniform porosity at the bearing surface and upon the prevention of contamination of the oil contained in the bearing.

3.3 Equipment.

3.3.1 Shouldered arbor plug. A shouldered arbor plug inserted in an arbor press shall be used to install and simultaneously size sintered metal bearings. The outer diameter of the plug shall be 0.0001 to 0.0003 inch larger (depending on the wall thickness of the bushing being installed) than the desired finished inner diameter of the installed bearing.

3.3.2 Cutting tool bits and drills. Only dead sharp diamond and carbide tipped cutting tools and drills shall be used for the rough and finish cuts on sintered metal bearing material.

3.3.3 Roller burnisher. A roller type burnisher used on a drill press set-up shall normally be used to size the inner diameter of installed bushings. A burnisher with an adjustment range of 0.002 and 0.037 inch above nominal hole size depending on tool size shall be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:25Z
Check the source to verify that this is the current version before use.

3.3.4 Reamers. A reamer may be used to size the inner diameter of the bushing after installation. The cutting edge of the reamer shall be dead sharp.

3.4 Machining special bushings or parts.

3.4.1 Only cutting tools in accordance with 3.3.2 shall be used in machining operations. No coolants or lubricants shall be used. For finish cuts fast speeds and fine feeds shall be used. The finish cuts shall not exceed 0.002 inch.

* 3.4.2 The finished parts shall be free from tool marks, nicks, cracks or defects which would affect their serviceability. The surface roughness shall not exceed RMS 63 when determined as prescribed in ANSI B46.1.

* 3.4.3 After final machining the bushings shall be cleaned with forced air and, except as stated in 3.4.3.1, reoiled by immersing in oil conforming to VV-L-800 for a minimum of 20 minutes at 140°F.

3.4.3.1 After final machining bushings made from iron base sintered material (except stainless steel) shall be cadmium plated as specified in 3.5 prior to relubrication in accordance with 3.4.3.

* 3.4.4 After relubrication the bushings shall be wrapped in barrier material conforming to MIL-B-121 or in other suitable nonabsorbent material such as wax paper or cellophane prior to handling or storage.

3.5 Cadmium plating and relubrication of iron base material. After final machining iron base self-lubricating bearings shall be degreased, plated, treated and relubricated in accordance with the following subparagraphs.

3.5.1 Preparation before plating. The following procedure shall be followed prior to the plating of oil impregnated iron base bearings:

a. Remove oil and grease by vapor degreasing with organic solvents of trichlorethylene type, clean lead-free gasoline or equivalent.

b. Immerse in a hot alkaline bath as used in best commercial practice.

c. Rinse thoroughly in water, using 2 rinses.

d. Dip in a 10 percent hydrochloric acid bath for 10 to 20 seconds.

e. Rinse thoroughly in cold water, followed by a rinse in hot water.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:25Z
Check the source to verify that this is the current version before use.

3.5.2 Cadmium plating. Immediately following the degreasing operation described above, iron base bushings shall be cadmium plated in accordance with QQ-P-416. The plating shall be type I and the minimum thickness requirement shall be class 3 (0.0002 inch) of QQ-P-416.

3.5.2.1 Adherence of the plating shall be such that the coating examined at a magnification of approximately 4 diameters does not show separation from the base metal when the surface of the plated part is scraped to the base metal.

3.5.3 After completion of the plating process, the bearings shall be thoroughly rinsed in cold water, followed by a hot water rinse. The bearings shall then be treated as follows to remove all traces of cyanide salts from the pl ting solution:

a. Wash each individual bearing with water under pressure of 60 psi for 30 to 60 seconds.

b. Force air (dry) under pressure of 100 psi through each individual bearing to remove water.

c. Repeat the washing and drying operations described in a and b 4 times.

d. Dry the bearings in an oven at 250°F for 1/2 to 1 hour.

3.5.4 Relubrication. The plated bearings shall be relubricated in accordance with 3.4.3. The reoilded bearings shall be protected as specified in 3.4.4.

3.5.5 When bearings are procured in the unimpregnated condition 3.5.1 may be disregarded.

3.6 Installation of bearings.

3.6.1 Preparation for installation. Prior to installing any bushing, both the installation hole and the bushing shall be inspected for conformance to dimensional requirements in accordance with detailed drawings or specifications. The housing hole shall be chambered to 1/64 by 45°, or greater when specified on the applicable drawing. Both the housing hole and the bushing shall be clean and free from chips or slivers. No lubricant, grease, or paint shall be used on the mating surfaces (lubrication is provided from the bushing).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:25Z
Check the source to verify that this is the current version before use.

3.6.2 Installation. The bearing shall be pressed into the housing with a shouldered arbor plug or other suitable means. The pressing load applied shall be parallel to the bushing centerline. High impact forces shall not be used.

3.6.3 After installation and prior to sizing the inner diameter shall be checked to determine that the material to be burnished or removed is not excessive (see 3.6.4.2 and 3.6.4.3).

3.6.4 When a sizing mandrel in accordance with 3.3.1 is not used for simultaneous installation and sizing, the bushing may be sized after installation as described in the following subparagraphs.

3.6.4.1 Bushings over 1 inch inner diameter shall be jig bored, radial drilled or turned to final size.

3.6.4.2 Bushings of 1 inch or smaller inner diameter shall normally be sized by burnishing with a roller type burnisher (see 3.3.3). The bushing inner diameter shall be not more than 0.002 inch undersize prior to sizing by this method.

3.6.4.3 Bushings of 1 inch or smaller inner diameter may be reamed to final size. Reamers in accordance wth 3.3.4 shall be used. No more than 0.0015 inch of material shall be removed during this sizing operation.

4. QUALITY ASSURANCE RROVISIONS

4.1 Inspection and evaluation shall be conducted to the extent necessary to determine and insure compliance with the requirements and provisions of this specification.

4.2 Tests, when required, on the sintered material, the lubricant or the plating process shall be in accordance with methods outlined in the applicable specification.

* 4.3 Unless otherwise specified, tests on self-lubricating bearings produced under this specification to assure conformance to 3.1.3 shall be in accordance with MIL-B-5687.

4.4 Tests on bushings which have been reworked, plated or relubricated shall be conducted in accordance with the methods outlined in the applicable specification to determine that the porosity and self-lubricating properties of the inner diameter have not been impaired.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:25Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 There are no applicable requirements.

6. NOTES

6.1 Intended use. The process and installation requirements in this specification are intended for use in the machining, sizing, degreasing, cadmium plating, and relubrication of oil impregnated sintered metal material for use as bushing type bearings in aircraft applications.

6.2 Marginal indicia. Asterisks(*) are used in this revision to identify changes (additions, deletions) with respect to the previous issued.

Custodian:
Air Force - 11

Reviewers:
Air Force - 13, 84
DSA - SG

Preparing activity:
Air Force - 11

Project No. 3120-F403

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T07:25Z
Check the source to verify that this is the current version before use.